CRF Simulator Canonical Specification

Grounding Every Simulation Variable in the δ-Chain

Reference document for all simulator branches

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Abstract

This document establishes the canonical mapping between CRF formalism and simulator variables across all branches (GPT, Claude, Gemini). Every variable used in simulation must derive from the δ-chain. This document identifies what is currently valid, what is approximated, what is broken, and what is missing. It is the ground truth for forward development.

Contents

1 The Immovable Chain

All simulator variables must trace back to this chain. No exceptions.

The δ-Chain (Status: Closed)

$$\delta \to P \to C \equiv \Pr(s|C) \to D_{\mathrm{KL}} \to \theta = 1 - e^{-D_{\mathrm{KL}}/D_0} \to R \to t = |R_{\text{events}}|$$

Physical constants derived without free parameters:

$$\alpha = \ln 2, \quad \kappa = \frac{k_B \ln 2}{H_{\text{mat}}}, \quad G = \frac{\ln(2) \cdot c^2}{D_0}, \quad \varepsilon_0 = |I_{\text{eff}} - \ln 2| \approx 0.00755$$

2 Canonical Variable Mapping

2.1 Variables That Are Correctly Implemented

Valid Simulator Variables

CRF Symbol	CRF Definition	Simulator Implementation
δ	Pre-primitive, self-grounding	R-event trigger: tension > threshold
P	$\delta(\emptyset)$, possibility space	`P[i]`: probability vector, simplex
D_{KL}	$\sum p \log(p/q)$, info gap	`KL(P[i], P[j])`: pairwise
$\alpha = \ln 2$	δ-bit equivalence	`self.ln2 = np.log(2)`
ε_0	$\lvert I_{\text{eff}} - \ln 2\rvert$ instability floor	`self.epsilon_0 = 0.00755`
$\Omega \equiv D_{\mathrm{KL}}$	EIC field = CRF gap	`KL(P[i], P[j])`
R-event	$\Pr(s) \geq \theta$ crossing	tension > local_threshold
$t = \lvert R_{\text{events}}\rvert$	Time from resolution count	`local_time[i] += 1`
Born Rule	$\lvert\langle s\|\psi\rangle\rvert^2$ at $C_{\min}$	`born_rule_resolution()`

2.2 Variables That Are Approximated

Approximated — Acceptable but Should Be Noted

CRF Symbol	CRF Definition	Current Approximation
$C \equiv \Pr(s\|C)$	Self-referential constraint	`node_mass` $= 1+$ `local_time` $\times \kappa$
		Correct direction, not self-referential loop
N_{eff}	$\sum_i \Delta D_{\text{KL},i}$	`local_time[i]`: count of R-events
		Should weight by ΔD_{KL}, not just count
θ	$1 - e^{-D_{\text{KL}}/D_0}$	`R_THRESHOLD = 0.015` (fixed scalar)
		Should be node-local, computed from D_{KL}

2.3 Variables That Are Broken

Broken — Must Be Fixed Before Building Further

CRF Symbol	What Should Happen	Current Broken Form
θ_{node}	$1 - e^{-D_{\text{KL,local}}/D_0}$	Fixed constant 0.015
	Dynamic, D_{KL}-dependent	Loses all dynamic threshold behavior
Gravity probe	$D_{\text{KL}}(P_i, P_j) \propto m_i m_j / r^2$	V15: pull = KL $\times r$
	KL alone is the coupling	Extra r inverts the correlation
d_s estimator	Single consistent method	3 different methods across versions
	Random-walk return slope	Matrix power / neighbor dist / walk

2.4 Variables That Are Missing

Missing — Not Yet Implemented in Any Branch

CRF Quantity	What Is Missing
$G = \ln(2) \cdot c^2/D_0$	Gravitational constant never extracted from simulation. G_{sim} should emerge from log-log slope intercept of $D_{\text{KL}} \propto m_1 m_2/r^2$
$\alpha_c \equiv \theta$	Critical threshold should be computed as $1 - e^{-\Omega_{\text{mean}}/D_0}$, not hardcoded
$\tau = \tau_0 e^{-\kappa N_{\text{eff}}}$	Collapse time never measured per node. This is the P0 prediction
$d_s \to 3$ proof	SO(3) minimality: formal proof that $n = 3$ is unique stable dimension. Simulation sees $d_s \approx 3$ but cannot prove it

3 The One Formula That Gravity Measurement Requires

Gravity Probe — Canonical Form

From CRF derivation chain:

$$C(r) = \frac{\alpha}{1 - r_s/r} \implies D_{\text{KL}} \approx \frac{C^2}{\alpha^2} \approx \frac{GM}{c^2 r} \quad \text{(weak field)}$$

Therefore the correct simulator measurement is:

$$\text{probe} = D_{\text{KL}}(P_i, P_j) \quad \text{plotted against} \quad \frac{m_i m_j}{r^2}$$

Not KL $\times r$. Not KL / r. Just D_{KL} directly.
If $D_{\text{KL}} \propto m_i m_j/r^2$, the log-log slope should be $+1$. The intercept gives $G_{\text{sim}} = \ln(2) \cdot c^2/D_0$.

4 The One Formula That θ Requires

Dynamic Threshold — Canonical Form

θ is not a free parameter. It derives from D_{KL}:

$$\theta_i = 1 - e^{-D_{\mathrm{KL,local},i}/D_0}$$

where $D_{\mathrm{KL,local},i} = \frac{1}{k}\sum_{j\in\mathrm{neighbors}(i)} D_{\mathrm{KL}}(P_i, P_j)$.
In simulator code this must be computed per node per sweep, not set as a global constant. When $D_{\mathrm{KL,local}} \to 0$: $\theta \to 0$, R-events become rare (crystallized state). When $D_{\mathrm{KL,local}}$ is high: θ is high, system needs stronger signal to resolve. This is the self-regulating mechanism that produces $\alpha_c = 0.525$.

5 What Is Valid to Build On Right Now

Solid Foundation — Build From Here

The following chain in the simulator is correctly grounded in CRF:

1. **P-space with KL neighbors**: nodes linked by D_{KL} is correct. D_{KL} is the natural distance in CRF's information geometry.
2. **R-event as Born rule resolution**: `born_rule_resolution()` using Dirichlet sampling anchored at ε_0 is the correct implementation of $\Pr(s|C_{\min}) = |\langle s|\psi\rangle|^2$.
3. **local_time as $t = |R_{\mathrm{events}}|$**: each node's clock is its R-event count. This is exactly $t = |R_{\mathrm{events}}|$ in CRF.
4. **mass from history**: `node_mass` $= 1+$ `local_time` $\times\kappa$. The direction is correct: more R-events $\implies$ more constraint $\implies$ more mass. This is $C \propto N_{\mathrm{eff}}$.
5. **ε_0 as noise floor**: using $\varepsilon_0 = 0.00755$ as the minimum fluctuation in every stochastic step is the correct implementation of the instability floor from EIC SINDy.
6. **Core detection by mass**: top nodes by `local_time` are nodes with the most R-event history. These are the high-C excitations. Correct.

6 The Minimum Fix for V16

Three Changes That Make V16 CRF-Correct

Fix 1: Dynamic θ
Replace `R_THRESHOLD = 0.015` with per-node computation:

$$\theta_i = 1 - \exp\left(-\frac{\bar{D}_{\mathrm{KL},i}}{D_0}\right), \quad \bar{D}_{\mathrm{KL},i} = \frac{1}{k}\sum_j D_{\mathrm{KL}}(P_i, P_j)$$

Fix 2: Gravity probe without r
Replace `pull = KL(P_i, P_j) * r` with:

$$\text{coupling} = D_{\mathrm{KL}}(P_i, P_j)$$

Plot against $m_i m_j / r^2$. Slope +1 confirms Newton. Intercept gives G_{sim}.
Fix 3: Consistent d_s estimator
Use random-walk return method only:

$$d_s = -2 \cdot \frac{\log p_{\mathrm{return}}(t)}{\log t}, \quad p_{\mathrm{return}}(t) = \frac{|\{w : w_t = w_0\}|}{N_{\mathrm{walkers}}}$$

Same formula in every version.

7 Status Table

Component	Status	Note
P-space with KL topology	Derived	Correct in all branches
Born rule resolution	Derived	Correct in V12+
$t = \lvert R_{\mathrm{events}} \rvert$	Derived	Correct in all branches
mass $\propto N_{\mathrm{eff}}$	Derived	Direction correct
ε_0 as noise floor	Confirmed	EIC SINDy verified
θ as dynamic per-node	Open	Fixed constant in all versions
Gravity probe $= D_{\mathrm{KL}}$ only	Open	V15 adds spurious $\times r$
G_{sim} from intercept	Hypothesis	V14.6 R^2=0.817 promising
$d_s \to 3$ proof	Open	Seen but not proved
$\tau = \tau_0 e^{-\kappa N_{\mathrm{eff}}}$	Open	Never measured

The simulator does not need to be rebuilt. It needs three precise corrections, each grounded in the same chain: $\delta \to D_{\mathrm{KL}} \to \theta \to R \to t$.